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A new intuitionistic fuzzy correlation coefficient approach with applications in multi-criteria decision-making

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ABSTRACT

The Intuitionistic Fuzzy Set (IFS) is a powerful extension of Fuzzy Set (FS) used to handle vagueness and uncertainty. The correlation coefficient is an established tool for finding the strength of relationships between two $IFSs$. This study proposes a new approach for finding a correlation coefficient to compare with the existing approach and show its superiority over the existing approach. In many counter-intuitive cases, existing approaches fail to handle the problem appropriately by giving indeterminate form, incorrect and inaccurate results. We demonstrate the reliability and efficiency of the proposed approach in handling these concerns. We also present the application of the proposed approach in pattern detection, medical illness diagnosis, clustering, and Multi-Criteria Decision-Making ($MCDM$) to further exhibit its practicability and effectiveness.

1. Introduction

In 1965, a powerful concept of fuzzy set (FS) (different from the classical set) was introduced by Zadeh [1], which mainly took the concept of degree of membership (DoM) of an element of a set. Different kinds of phenomena were considered and studied under this concept. But some phenomena still go untouched. To cover a new phenomenon, a generalization intuitionistic fuzzy set (IFS) of FSs was proposed by Atanassov [2]. It includes the DoM as well as the degree of non-membership ($DoNM$) of elements. So, some disadvantages of FSs were overcome by $IFSs$. After that, for more applications Atanassov [3] extended IFS in the form of interval-valued intuitionistic fuzzy set ($IvIFS$), DoM and $DoNM$ were taken as intervals. All these topics were widely applied in different fields like agriculture, diagnosis of medical illness, decision-making problems, agriculture, cluster analysis, and education area, which were discussed in [4–12]. Furthermore, research in the field of $IFSs$ can be mainly divided into four concepts: aggregation operators, preference relation, decision-making methods, and measures. A geometric aggregation operator, a weighted averaging operator, and a generalized intuitionistic fuzzy interactive geometric aggregation operator (using Einstein t-norm and t-conorm) were proposed in [13–15] respectively. Similar methods for $IvIFSs$ were developed in [16,17].

In recent years, the correlation coefficient (CC) for FS has been an eye-catching topic for solving decision-making issues in real life and cluster analysis, etc. It is a very useful tool to find the strength of the relationship between different variables. This topic has value in the field of statistics and engineering. Researchers are adopting new methods for creating effective formulas of CC. In the study of the research of past years, it is found that the CC can take values in [0,1]. Also, in the study of recent years, it can be clearly observed that the value of CC can lie in [-1,-1]. A formula for finding a correlation between FSs, based on traditional statistics was developed in [18]. After that, a new formula for correlation between FSs, by using mathematical programming was studied in [19]. Furthermore, Gerstenkorn and Manko [20] were the first who studied the CC of $IFSs$. But, in this formula, the hesitation index was neglected, so it was unable to produce the correct output. To overcome this limitation, a previous approach was considered and a new formula was derived in [21]. This was also not up to the mark, it gave no correlation between many $IFSs$, which was again wrong output. Then, a new directional CC based on four parameters namely, DoM, $DoNM$, the strength of commitment, and direction of commitment of Pythagorean fuzzy sets is proposed in [22]. After this new approach of CC, again a new method of calculating the CC for interval valued intuitionistic fuzzy soft sets is suggested in [23] to solve decision making issues. Furthermore, Mingwei Lin [24] put forward correlation coefficient formulas to measure the relationship between probabilistic linguistic sets and applied them for clustering. After that, the formula of CC, based on the set theory viewpoint was defined in [25]. All the formulae did not give accurate output, most of them gave either no correlation or perfect correlation as output for many $IFSs$, which are discussed in this paper. Furthermore, Xu and Huang [26,27], and [28] developed their formulae of correlation to overcome the flaws of existing formulae, but all of them were not reliable. Somehow, they were unable to yield the correct output, for many counter-examples.

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Abbreviations

FS	Fuzzy Set
IF	Intuitionistic fuzzy
IFS	Intuitionistic fuzzy set
CC	Correlation coefficient
DoM	Degree of membership
$DoNM$	Degree of non-membership
VIKOR	Vlekriterijumsko KOmpromisno Rangiranje
TOPSIS	Technique for Order Performance by Similarity to Ideal Solution
$MCDM$	Multi criteria decision-making
$IVIFS$	Interval valued intuitionistic fuzzy set

In this paper, an improved and modified approach is proposed for finding the CC between $IFSs$, to overcome the drawbacks of existing approaches. This new approach has been applied in many fields to help in solving decision-making issues related to real life. Pattern detection is an art, that is related to the grouping and categorization of the given data. A pattern detection issue was discussed in [29] based on similarity measures, furthermore, a two-parametric similarity measure for $IFSs$ was discussed by Boran and Akay [30] and applied it pattern detection. In the same way, pattern detection was studied in [31,32], which was based on a distance measure. In this paper, the approach of a correlation coefficient is effectively applied for pattern detection with the help of illustrative examples. Furthermore, the application of similarity measure, dissimilarity measure and CC in the field of medical diagnosis were discussed in [6,7,28,33], and [34]. Nowadays, clustering is a well-known and helpful tool in solving problems related to the grouping of the data in FSs. Clustering can be defined as an act of segregating the unlabeled data into different clusters such that similar data come in the same cluster. In simple words, the clustering process aims to segregate groups with similar properties and assign them into clusters. The application of the CC for $IFSs$ in solving the issues regarding cluster analysis were discussed in [28,35], and [36]. There is great significance in handling the problems of multi-criteria decision-making ($MCDM$) issues in an intuitionistic fuzzy environment. Different methods have been developed over the years in the field of solving decision-making issues. $MCDM$ issues are handled in [37] for interval valued pythagorean fuzzy sets. In [13], Xu applied some geometric, weighted geometric, and hybrid geometric operators in the $IFSs$ environment to solve multi-attribute decision-making issues ($MADM$). Mingwei Lin [38] used correlation coefficient in Technique for Order of Preference by Similarity to the Ideal Solution (TOPSIS) method to handle $MADM$ issues, Ejegwa [39] used pythagorean fuzzy CC approach based on variance and covariance to solve $MADM$ issues, and Chakraborty [40] used modified TOPSIS method to solve $MADM$ issues. Zhang [41] used different methods like the information entropy method to determine the attribute weight, the combined grey relational analysis method, the grey relational projection method, and the proposed projection method for decision-making issues. After studying the advantages of the Vlekriterijumsko KOmpromisno Rangiranje (VIKOR) method developed in [42], researchers got a new topic for their study in solving decision-making issues. Tian [43] discussed the extended VIKOR method which was based on the similarity measure of the picture fuzzy set, but it considered only those values, which were close to the positive ideal value. The method proposed in [43] was modified in [44] and a new improved VIKOR method based on hybrid similarity measure in a Picture fuzzy environment was developed, which overcame the flaw of the previous method. A new decision-making method was also proposed by Wang and Gong based on set pair analysis. After that Cao [45] developed a new approach using set pair analysis, a similarity measure to deal with $MCDM$ issues in $IvIFSs$. Furthermore, CC was used for solving $MCDM$ issues in $IFSs$ environment in [46]. A new triangular fuzzy $MCDM$ method based on set pair analysis was developed in [47]. Again, a (TOPSIS) in the $IvIFSs$ environment was derived in [48]. Some more researchers [12,49–55] have also solved decision-making problems.

Therefore, after observing the previous research, in this paper, we have developed a new approach to finding the CC for $IFSs$. This new approach is better than the existing approaches in the following manner:

- Unlike many of the existing approaches, it includes the hesitancy index, which makes this approach more reliable.
- It gives 0 value only when there is no correlation between $IFSs$.
- It gives 1 value only when there is a perfect correlation between $IFSs$, that is when the sets are equal.
- Unlike many of the existing approaches, it does not exceed its value of 1.
- There is no case of $\frac{0}{0}$ forms in this approach.
- Many of the existing approaches are not applicable to singleton sets. But, it can deal with these types of sets.
- The rate of performance and accuracy in calculations is higher than that of others.

Also, this approach can be applied to handle problems related to real life like pattern detection problems, diagnosing medical illness, clustering problems, and $MCDM$ problems. So, this approach is highly effective, reliable, accurate in giving results, and applicable in many decision-making issues. This paper has the following structure: Section 2 describes the basic definitions to make it easy to understand the paper, Section 3 illustrates the existing approaches of CC with numerical examples showing their shortcomings, Section 4 comprises the new proposed approach with some basic properties and comparative analysis with existing approaches, Section 5 includes its applications in Pattern detection, medical diagnosis, clustering and $MCDM$ issues with numerical examples and results comparison with existing approaches. At last, this paper is concluded in Section 6 with an overall short review of the work and future applications.

2. Preliminaries

In this section, we shall provide some basic definitions and concepts explained as follows:

Definition 1 ([1]). Let $\ddot{U} = \{u_1, u_2, \ldots, u_n\}$ be finite set. Then, an FS $\check{F}$ in $\ddot{U}$ can be defined as

$$\check{F} = \{(u, m_{\check{F}}(u)) \eth u \in \ddot{U}\},$$

where $m_{\check{F}} : \ddot{U} \to [0,1]$ establishes a function for DoM of u in set $\check{F}$.

Definition 2 ([56,57]). Let $\ddot{U} = \{u_1, u_2, \ldots, u_n\}$ be a finite set. Then, an IFS $\check{F}$ in $\ddot{U}$ can be defined as

$$\check{F} = \{(u, m_{\check{F}}(u), n_{\check{F}}(u), \rho_{\check{F}}(u)) \eth u \in \ddot{U}\},$$

where $m_{\check{F}} : \ddot{U} \to [0,1]$ establishes a function for DoM of u in $\check{F}$ and $n_{\check{F}} : \ddot{U} \to [0,1]$ establishes a function for $DoNM$ of u in $\check{F}$ such that $0 \leq m_{\check{F}}(u) + n_{\check{F}}(u) \leq 1$ for all $u \in \ddot{U}$. Hesitancy index is $\rho_{\check{F}}(u) = 1 - (m_{\check{F}}(u) + n_{\check{F}}(u))$.

Definition 3 ([2]). Some Operations on $IFSs$ are defined as follows:

Let $IFS(\ddot{U})$ be the set containing all the $IFSs$ defined on $\ddot{U} = \{u_1, u_2, \ldots, u_n\}$. The operations given below are applicable for all $\check{F}, \check{G} \in IFS(\ddot{U})$:

(i) $\check{F} = \check{G}$ iff $m_{\check{F}}(u) = m_{\check{G}}(u)$ and $n_{\check{F}}(u) = n_{\check{G}}(u)$, $\forall u \in \ddot{U}$,

(ii) $\check{F} \subseteq \check{G}$ iff $m_{\check{F}}(u) \leq m_{\check{G}}(u)$ and $n_{\check{F}}(u) \geq n_{\check{G}}(u)$, $\forall u \in \ddot{U}$,

$(iii)\ \check{F}^C = \{(u, n_{\check{F}}(u),\ m_{\check{F}}(u))\text{ð}\ u \in \ddot{U}\},$

$(iv)\ \check{F} \cup \check{G} = \{(max\ \{m_{\check{F}}(u), m_{\check{G}}(u)\},\ min\ \{n_{\check{F}}(u), n_{\check{G}}(u)\})\text{ð}\ u \in \ddot{U}\},$

$(v)\ \check{F} \cap \check{G} = \{(min\ \{m_{\check{F}}(u), m_{\check{G}}(u)\},\ max\ \{n_{\check{F}}(u), n_{\check{G}}(u)\})\text{ð}\ u \in \ddot{U}\}.$

Definition 4 ([20]). Let $I\,FS(\ddot{U})$ be the set containing all the $IFSs$ defined on $\ddot{U} = \{u_1, u_2, \ldots, u_n\}$ and $\check{F}, \check{G} \in I\,FS(\ddot{U})$. Then, the coefficient of correlation between $\check{F}$ and $\check{G}$ with notation $\tau(\check{F}, \check{G})$ is a function

$\tau : I\,FS(\ddot{U}) \times I\,FS(\ddot{U})$ ™[0,1] such that the following conditions hold:

1. 0 f $\tau(\check{F}, \check{G})$ f 1;
2. $\tau(\check{F}, \check{G}) = 1$ if and only if $\check{F} = \check{G}$;
3. $\tau(\check{F}, \check{G}) = \tau(\check{F}, \check{G})$.

$\tau(\check{F}, \check{G})$ approaches 1 implies that $\check{F}$ and $\check{G}$ have a strong correlation. Also, $\tau(\check{F}, \check{G})$ approaches 0 implies that $\check{F}$ and $\check{G}$ have a weak correlation. But, $\tau(\check{F}, \check{G}) = 0$ implies that $\check{F}$ and $\check{G}$ do not correlate. Therefore, a bigger correlation coefficient represents a better rate of performance.

Definition 5 ([25]). Let $\check{Q}_i \in I\,FS(\ddot{U})$, $i = 1, 2, \ldots, r$, then $\tilde{Y} = [y_{ij}]_{r\times r}$ is considered as an association matrix, where $y_{ij} = \tau_J(\check{Q}_i, \check{Q}_j)$, $(i, j = 1, 2, \ldots, r)$ is the CC of $\check{Q}_i$ and $\check{Q}_j$, having following properties:

(i) 0 f y_{ij} f 1 $(i, j = 1, 2, \ldots, r)$;
(ii) $y_{ij} = y_{ji}$ $(i, j = 1, 2, \ldots, r)$;
(iii) $y_{ij} = 1$ if and only if $\check{Q}_i = \check{Q}_j$.

Definition 6 ([25]). Let $\tilde{Y} = [y_{ij}]_{r\times r}$ be a association matrix, then $\tilde{Y}^2 = \tilde{Y}\acute{y}\tilde{Y} = [\bar{y_{ij}}]$ is considered as composition matrix of $\tilde{Y}$, where $\bar{y_{ij}} = \max_k\{\min\{y_{ik}, y_{kj}\}\}$ and $(i, j = 1, 2, \ldots, r)$.

Definition 7 ([25]). Let $\tilde{Y} = [y_{ij}]_{r\times r}$ be an equivalent association matrix, then $\tilde{Y}_\lambda = [{}_\lambda y_{ij}]_{r\times r}$ is considered as λ- cutting matrix of $\tilde{Y}$,where ${}_\lambda y_{ij} = 0$ for y_{ij} f λ and ${}_\lambda y_{ij} = 1$ for y_{ij} g λ. λ is the confidence level such that $\lambda \in [0, 1]$.

Definition 8 ([58]). An entropy on $I\,FS(\ddot{U})$ can be defined as $I : I\,FS(\ddot{U})$ ™[0,1] such that the following conditions hold:

(i) $I(\check{F})$=0 if $\check{F}$ is a crisp set,
(ii) $I(\check{F})$=1 if $\check{F} = \{(u_i, \frac{1}{3}, \frac{1}{3})\text{ð}u_i \in \ddot{U}\}$,
(iii) $I(\check{F}) = I(\check{F}^c)$, $\forall \check{F} \in I\,FS(\ddot{U})$.
(iv) $I(\check{F})$ f $I(\check{G})$, $\forall \check{F}, \check{G} \in I\,FS(\ddot{U})$ satisfying:

either $m_{\check{F}}(u_i)$ f $m_{\check{G}}(u_i), n_{\check{F}}(u_i)$ f $n_{\check{G}}(u_i)$, if $max\ \{m_{\check{G}}(u_i), m_{\check{G}}(u_i)\}$ f $\frac{1}{3}$

or $m_{\check{F}}(u_i)$ g $m_{\check{G}}(u_i), n_{\check{F}}(u_i)$ g $n_{\check{G}}(u_i)$, if $min\ \{m_{\check{F}}(u_i), m_{\check{F}}(u_i)\}$ g $\frac{1}{3}$, $\forall\, u_i \in (\ddot{U})$.

3. Existing correlation coefficient measures

In this section, we use the existing CC measures of IFS for comparison.

3.1. CC In [20]

$$\tau_1(\check{F}, \check{G}) = \frac{\mathsf{C}(\check{F}, \check{G})}{\sqrt{\mathsf{T}(\check{F})}\sqrt{\mathsf{T}(\check{G})}}, \text{where} \tag{1}$$

$$\begin{aligned}\mathsf{C}(\check{F}, \check{G}) &= \sum_{i=1}^{n}[m_{\check{F}}(u_i)m_{\check{G}}(u_i) + n_{\check{F}}(u_i)n_{\check{G}}(u_i)],\\ \mathsf{T}(\check{F}) &= \sum_{i=1}^{n}[m^2_{\check{F}}(u_i) + n^2_{\check{F}}(u_i)],\\ \mathsf{T}(\check{G}) &= \sum_{i=1}^{n}[m^2_{\check{G}}(u_i) + n^2_{\check{G}}(u_i)].\end{aligned} \tag{2}$$

It can be observed that the hesitation index is not included in the above formula for the CC of $IFSs$, so its output cannot be trusted.

Example 1. Let $\check{F}$ and $\check{G}$ be two $IFSs$ given by
$\check{F} = \{(u_1, 1, 0), (u_2, 0, 0.5)\}$ and $\check{G} = \{(u_1, 0, 0.5), (u_2, 1, 0)\}$, where $\ddot{U} = \{u_1, u_2\}$.

By applying above formula, we get $\mathsf{C}(\check{F}, \check{G}) = 0, \mathsf{T}(F) = 1.25 = \mathsf{T}(G)$ and

$$\tau_1(\check{F}, \check{G}) = \frac{0}{\sqrt{(1.25)}\sqrt{(1.25)}} = 0.$$

This gives the wrong output for the correlation between $\check{F}$ and $\check{G}$.

Example 2. Let $\check{F}$ and $\check{G}$ be two $IFSs$ given by
$\check{F} = \{(u_1, 0.135, 0.635), (u_2, 0.035, 0.820), (u_3, 0.050, 0.700), (u_4, 0.200, 0.575), (u_5, 0.115, 0.815)\}$ and
$\check{G} = \{(u_1, 0.529, 0.203), (u_2, 0.526, 0.334), (u_3, 0.210, 0.675), (u_4, 0.035, 0.555), (u_5, 0.538, 0.432)\}$, where $\ddot{U} = \{u_1, u_2, u_3, u_4, u_5\}$.

By applying above formula, we get $\mathsf{C}(\check{F}, \check{G}) = 1.716$, $\sqrt{\mathsf{T}(\check{F})} = 1.057$, $\sqrt{\mathsf{T}(\check{G})} = 1.42$ and

$$\tau_1(\check{F}, \check{G}) = \frac{1.716}{(1.057)(1.42)} = 1.143 > 1.$$

Clearly, this gives an output for correlation between $\check{F}$ and $\check{G}$, which is not acceptable.

3.2. CC In [21]

$$\tau_2(\check{F}, \check{G}) = \frac{\mathsf{C}(\check{F}, \check{G})}{\sqrt{\mathsf{T}(\check{F})}\sqrt{\mathsf{T}(\check{G})}}, \text{where} \tag{3}$$

$$\begin{aligned}\mathsf{C}(\check{F}, \check{G}) &= \frac{\sqrt[3]{\sum_{i=1}^{N}[m_{\check{F}}(u_i)m_{\check{G}}(u_i) + n_{\check{F}}(u_i)n_{\check{G}}(u_i) + \rho_{\check{F}}(u_i)\rho_{\check{G}}(u_i)]}}{N},\\ \mathsf{T}(\check{F}) &= \frac{\sqrt[3]{\sum_{i=1}^{N}[m^2_{\check{F}}(u_i) + n^2_{\check{F}}(u_i) + \rho^2_{\check{F}}(u_i)]}}{N},\\ \mathsf{T}(\check{G}) &= \frac{\sqrt[3]{\sum_{i=1}^{N}[m^2_{\check{G}}(u_i) + n^2_{\check{G}}(u_i) + \rho^2_{\check{G}}(u_i)]}}{N}.\end{aligned} \tag{4}$$

If we apply this formula to the same Example 1, we get $\mathsf{C}(\check{F}, \check{G}) = 0$, $\mathsf{T}(\check{F}) = \frac{1.50}{2} = 0.75 = \mathsf{T}(\check{G})$ and

$$\tau_2(\check{F}, \check{G}) = \frac{0}{\sqrt{(0.75)}\sqrt{(0.75)}} = 0.$$

Again, this formula also gives the wrong output regarding the correlation between $\check{F}$ and $\check{G}$. Therefore, this approach cannot be accepted.

3.3. CC In [25]

$$\tau_3(\check{F}, \check{G}) = \frac{\mathsf{C}(\check{F}, \check{G})}{\max\{\sqrt{\mathsf{T}(\check{F})}, \sqrt{\mathsf{T}(\check{G})}\}}, \text{where} \tag{5}$$

$$\begin{aligned} C(\check{F},\check{G}) &= \sum_{i=1}^{n} [m_{\check{F}}(u_i)m_{\check{G}}(u_i) + n_{\check{F}}(u_i)n_{\check{G}}(u_i)], \\ T(\check{F}) &= \sum_{i=1}^{n} [m^2_{\check{F}}(u_i) + n^2_{\check{F}}(u_i)], \\ T(\check{G}) &= \sum_{i=1}^{n} [m^2_{\check{G}}(u_i) + n^2_{\check{G}}(u_i)]. \end{aligned} \tag{6}$$

Here, in the above formula hesitation index is neglected. After applying this formula to the same Example 1, we get $C(\check{F},\check{G}) = 0, T(\check{F}) = 1.25 = T(\check{G})$ and

$$\tau_3(\check{F},\check{G}) = \frac{0}{\max\{\sqrt{(1.25)}, \sqrt{(1.25)}\}} = 0.$$

This is again a misinformation for the correlation coefficient between $\check{F}$ and $\check{G}$.

Also, when we apply this formula to Example 2, we get $C(\check{F},\check{G}) = 1.716$, $T(\check{F}) = 1.057$, $T(\check{G}) = 1.42$ and

$$\tau_3(\check{F},\check{G}) = \frac{1.716}{\max\{(1.057),(1.42)\}} = 1.208 > 1.$$

This output can never be accepted as $1.208 > 1$. This makes the formula unreliable.

Two other forms of [25], based on the approach given by [20] are as follows:

$$\tau_4(\check{F},\check{G}) = \frac{C(\check{F},\check{G})}{\max\{\sqrt{T(\check{F})}, \sqrt{T(\check{G})}\}}, \tag{7}$$

$$\tau_5(\check{F},\check{G}) = \frac{C(\check{F},\check{G})}{\sqrt{T(\check{F})}\sqrt{T(\check{G})}}, \text{where} \tag{8}$$

$$\begin{aligned} C(\check{F},\check{G}) &= \frac{\sqrt[3]{\sum_{i=1}^{N}[m_{\check{F}}(u_i)m_{\check{G}}(u_i) + n_{\check{F}}(u_i)n_{\check{G}}(u_i) + \rho_{\check{F}}(u_i)\rho_{\check{G}}(u_i)]}}{N}, \\ T(F) &= \frac{\sqrt[3]{\sum_{i=1}^{N}[m^2_{\check{F}}(u_i) + n^2_{\check{F}}(u_i) + \rho^2_{\check{F}}(u_i)]}}{N}, \\ T(G) &= \frac{\sqrt[3]{\sum_{i=1}^{N}[m^2_{\check{G}}(u_i) + n^2_{\check{G}}(u_i) + \rho^2_{\check{G}}(u_i)]}}{N}. \end{aligned} \tag{9}$$

When we apply this formula (7) to Example 1, we get $\tau_4(\check{F},\check{G}) = 0$. Also, when we apply this formula (7) to Example 2, we get $C(\check{F},\check{G}) = 1.919$, $\sqrt{T(\check{F})} = 1.682$, $\sqrt{T(\check{G})} = 1.505$ and

$$\tau_4(\check{F},\check{G}) = \frac{1.919}{\max\{(1.682),(1.505)\}} = 1.141 > 1.$$

This output can never be accepted as $1.141 > 1$. This makes the formula unreliable.

Example 3. Let $\check{F}$ and $\check{G}$ be two $IFSs$ given by

$\check{F} = \{(u_1, 0.58, 0.32)\}$ and $\check{G} = \{(u_1, 0.58, 0.32)\}$, where $\check{U} = \{u_1\}$. By applying above formula (7), we get $C(F,G) = 0.448$, $\sqrt{T(\check{F})} = 0.67 = \sqrt{T(\check{G})}$ and

$$\tau_4(\check{F},\check{G}) = \frac{0.448}{\max(0.67),(0.67)} = 0.67 \neq 1.$$

This gives the wrong output for the correlation between $\check{F}$ and $\check{G}$ as $\check{F} = \check{G}$ but, it shows that there is no perfect correlation between them.

By simple calculations, we can observe that after applying formula (8) to Example 1, we get $C(\check{F},\check{G}) = 0, T(\check{F}) = 1.50 = T(\check{G})$ and

$$\tau_5(\check{F},\check{G}) = \frac{0}{\sqrt{(1.50)}\sqrt{(1.50)}} = 0.$$

This gives the wrong output for the correlation between $\check{F}$ and $\check{G}$. Therefore, both approaches (7) and (8) do not give correct results, which is a huge drawback.

Table 1
Calculation process.

$\check{U}$	Δm_i	Δn_i
u_1	0.1	0.1
u_2	0.1	0.1

Table 2
Calculation process.

$\check{U}$	Δm_i	Δn_i	$\Delta \rho_i$
u_1	0.1	0.1	0.2
u_2	0.1	0.1	0.2

3.4. CC In [26]

$$\tau_6(\check{F},\check{G}) = \frac{1}{2N}\sum_{i=1}^{n} \frac{\Delta m_{min} + \Delta m_{max}}{\Delta m_i + \Delta m_{max}} + \frac{\Delta n_{min} + \Delta n_{max}}{\Delta n_i + \Delta n_{max}}, \tag{10}$$

$$\begin{aligned} \Delta m_i &= \eth m_{\check{F}}(u_i) - m_{\check{G}}(u_i)\eth, \quad \Delta n_i = \eth n_{\check{F}}(u_i) - n_{\check{G}}(u_i)\eth; \\ \Delta m_{min} &= \min_i \eth m_{\check{F}}(u_i) - m_{\check{G}}(u_i)\eth, \quad \Delta n_{min} = \min_i \eth n_{\check{F}}(u_i) - n_{\check{G}}(u_i)\eth; \\ \Delta m_{max} &= \max_i \eth m_{\check{F}}(u_i) - m_{\check{G}}(u_i)\eth, \quad \Delta n_{max} = \max_i \eth n_{\check{F}}(u_i) - n_{\check{G}}(u_i)\eth. \end{aligned} \tag{11}$$

This formula is also not reliable as it also ignores the hesitation index here. After applying formula (10) to Example 1, we get $\tau_6(\check{F},\check{G}) = 1$. Also, when we apply this formula to Example 3, it is unable to handle this case as it gives a $\frac{0}{0}$ form. This shows its inability to find the CC.

Example 4. Let $\check{F}$ and $\check{G}$ be two $IFSs$ given by

$\check{F} = \{(u_1, 0.5, 0.4), (u_2, 0.4, 0.3)\}$ and $\check{G} = \{(u_1, 0.4, 0.3), (u_2, 0.3, 0.2)\}$, where $\check{U} = \{u_1, u_2\}$. After computation, we get $\Delta m_{min} = \Delta n_{min} = \Delta m_{max} = \Delta n_{max} = 0.1$ (see Table 1).

Therefore, $\tau_6(\check{F},\check{G}) = 1$. It is clear according to the definition that the information obtained by the above formula is not correct as $\check{F} \neq \check{G}$ and it gives a perfect correlation between $\check{F}$ and $\check{G}$. So, this approach is also not reliable.

3.5. CC In [27]

$$\tau_7(\check{F},\check{G}) = \frac{1}{3N}\sum_{i=1}^{n} \frac{\Delta m_{min} + \Delta m_{max}}{\Delta m_i + \Delta m_{max}} + \frac{\Delta n_{min} + \Delta n_{max}}{\Delta n_i + \Delta n_{max}} + \frac{\Delta \rho_{min} + \Delta \rho_{max}}{\Delta \rho_i + \Delta \rho_{max}}, \tag{12}$$

$$\begin{aligned} \Delta m_i &= \eth m_{\check{F}}(u_i) - m_{\check{G}}(u_i)\eth, \quad \Delta n_i = \eth n_{\check{F}}(u_i) - n_{\check{G}}(u_i)\eth, \\ \Delta \rho_i &= \eth \rho_{\check{F}}(u_i) - \rho_{\check{G}}(u_i)\eth; \\ \Delta m_{min} &= \min_i \eth m_{\check{F}}(u_i) - m_{\check{G}}(u_i)\eth, \quad \Delta n_{min} = \min_i \eth n_{\check{F}}(u_i) - n_{\check{G}}(u_i)\eth, \\ \Delta \rho_{min} &= \min_i \eth \rho_{\check{F}}(u_i) - \rho_{\check{G}}(u_i)\eth; \\ \Delta m_{max} &= \max_i \eth m_{\check{F}}(u_i) - m_{\check{G}}(u_i)\eth, \quad \Delta n_{max} = \max_i \eth n_{\check{F}}(u_i) - n_{\check{G}}(u_i)\eth, \\ \Delta \rho_{max} &= \max_i \eth \rho_{\check{F}}(u_i) - \rho_{\check{G}}(u_i)\eth. \end{aligned} \tag{13}$$

After applying the above formula to Example 3, we get the $\frac{0}{0}$ form, which makes it clear that this formula (12) is also not suitable for finding CC for $IFSs$. Again, it is not reliable as it gives the following results after application to Example 4 (see Table 2):

$\Delta m_{min} = \Delta n_{min} = \Delta m_{max} = \Delta n_{max} = 0.1, \Delta \rho_{max} = \Delta \rho_{min} = 0.2$. Therefore, $\tau_7(\check{F},\check{G}) = 1$. According to the definition, it is clear that the information obtained by the above formula is not correct, as $\check{F} \neq \check{G}$.

Table 3
Calculation process (when $\alpha = 3$).

$\ddot{U}$	Δm_i	Δn_i	ϕ_i	φ_i
u_1	0.1	0.1	1.0	1.0
u_2	0.6	0.5	0.783	0.833

3.6. CC In [28]

$$\tau_8(\check{F},\check{G}) = \frac{1}{2N}\sum_{i=1}^{n}\phi_i(1-\Delta m_i)+\varphi_i(1-\Delta n_i)\ , \text{where} \tag{14}$$

$$\begin{aligned}
\phi_i &= \frac{\alpha-\Delta m_i-\Delta m_{max}}{\alpha-\Delta m_{min}-\Delta m_{max}},\ \varphi_i = \frac{\alpha-\Delta n_i-\Delta n_{max}}{\alpha-\Delta n_{min}-\Delta n_{max}};\ \text{for } \alpha>2 \text{ and}\\
\Delta m_i &= \delta m_{\check{F}}(u_i)-m_{\check{G}}(u_i)\delta,\ \Delta n_i = \delta n_{\check{F}}(u_i)-n_{\check{G}}(u_i)\delta;\\
\Delta m_{min} &= \min_i \delta m_{\check{F}}(u_i)-m_{\check{G}}(u_i)\delta,\ \Delta n_{min} = \min_i \delta n_{\check{F}}(u_i)-n_{\check{G}}(u_i)\delta;\\
\Delta m_{max} &= \max_i \delta m_{\check{F}}(u_i)-m_{\check{G}}(u_i)\delta,\ \Delta n_{max} = \max_i \delta n_{\check{F}}(u_i)-n_{\check{G}}(u_i)\delta.
\end{aligned} \tag{15}$$

The formula, which neglects the hesitation index can never provide accurate results.

Example 5. Let $\check{F}$ and $\check{G}$ be two $IFSs$ given by $\check{F} = \{(u_1,0.7,0.2),(u_2,0.8,0.1)\}$ and $\check{G} = \{(u_1,0.6,0.3),(u_2,0.2,0.6)\}$, where $\ddot{U} = \{u_1,u_2\}$. Applying the above formula here, we get (see Table 3)

$$\begin{aligned}
&\Delta m_{min}=0.1,\ \Delta n_{min}=0.1,\ \Delta m_{max}=0.6,\ \Delta n_{max}=0.5,\\
&\phi_1=\frac{3-0.1-0.6}{3-0.1-0.6}=1,\ \phi_2=\frac{3-0.6-0.6}{3-0.1-0.6}=0.783\\
&\varphi_1=\frac{3-0.1-0.5}{3-0.1-0.5}=1,\ \varphi_2=\frac{3-0.5-0.5}{3-0.1-0.5}=0.833.
\end{aligned}$$

Therefore, $\tau_8(\check{F},\check{G}) = 0.633$. Here, when the formula is applied on Example 4 it gives $\tau_8(\check{F},\check{G}) = 1$ for equal sets. Therefore, this approach is not very reliable.

4. New intuitionistic Fuzzy correlation coefficient

A new approach is developed for finding the CC between $IFSs$ as follows:

$$\tau_J(\check{F},\check{G}) = \frac{1}{3N}\sum_{i=1}^{n}\phi_i(1-\Delta m_i)+\varphi_i(1-\Delta n_i)+\psi_i(1-\Delta\rho_i)\ , \text{where} \tag{16}$$

$$\begin{aligned}
\phi_i &= \frac{\alpha-\Delta m_i-\Delta m_{max}}{\alpha-\Delta m_{min}-\Delta m_{max}},\\
\varphi_i &= \frac{\alpha-\Delta n_i-\Delta n_{max}}{\alpha-\Delta n_{min}-\Delta n_{max}},\\
\psi_i &= \frac{\alpha-\Delta\rho_i-\Delta\rho_{max}}{\alpha-\Delta\rho_{min}-\Delta\rho_{max}},\ \alpha>2 \text{ and finite};
\end{aligned} \tag{17}$$

$$\begin{aligned}
\Delta m_i &= \frac{\delta m_{\check{F}}(u_i)-m_{\check{G}}(u_i)\delta+\delta m^2_{\check{F}}(u_i)-m^2_{\check{G}}(u_i)\delta}{2},\\
\Delta n_i &= \frac{\delta n_{\check{F}}(u_i)-n_{\check{G}}(u_i)\delta+\delta n^2_{\check{F}}(u_i)-n^2_{\check{G}}(u_i)\delta}{2},\\
\Delta\rho_i &= \frac{\delta\rho_{\check{F}}(u_i)-\rho_{\check{G}}(u_i)\delta+\delta\rho^2_{\check{F}}(u_i)-\rho^2_{\check{G}}(u_i)\delta}{2};\\
\Delta m_{min} &= \min_i(\Delta m_i),\ \Delta n_{min}=\min_i(\Delta n_i),\ \Delta\rho_{min}=\min_i(\Delta\rho_i);\\
\Delta m_{max} &= \max_i(\Delta m_i),\ \Delta n_{max}=\max_i(\Delta n_i),\ \Delta\rho_{max}=\max_i(\Delta\rho_i).
\end{aligned} \tag{18}$$

Example 6. Let $\check{F}$ and $\check{G}$ be two $IFSs$ given by

Table 4
Calculation process (when $\alpha = 3$).

$\ddot{U}$	Δm_i	Δn_i	$\Delta\rho_i$
u_1	0.345	0.140	0.115
u_2	0.200	0.200	0.160
u_3	0.115	0.000	0.050

$\check{F} = \{(u_1,0.8,0.1,0.1),(u_2,0.6,0.2,0.2),(u_3,0.7,0.3,0.0)\}$ and $\check{G} = \{(u_1,0.5,0.3,0.2),(u_2,0.4,0.2,0.4),(u_3,0.6,0.3,0.1)\}$, where $\ddot{U} = \{u_1,u_2,u_3\}$. After applying a new approach to given sets, we get (see Table 4)

$$\begin{aligned}
&\Delta m_{min}=0.115,\ \Delta n_{min}=0.0,\ \Delta\rho_{min}=0.05;\\
&\Delta m_{max}=0.345,\ \Delta n_{max}=0.2,\ \Delta\rho_{max}=0.16;\\
&\tau_J(\check{F},\check{G}) = \frac{1}{9}\ 0.595+0.774+0.885+0.817+0.743\\
&\qquad +1+0.865+0.807+0.95\ =0.826
\end{aligned}$$

From Table 5, it is obvious that the performance rate of the newly developed CC approach is the highest among all the existing approaches for all values of α.

4.1. Comparative analysis

In this section, a comparative analysis is done to show the superiority of the new approach of finding CC between $IFSs$. A better comparison between existing approaches and the proposed approach is presented in the form of Table 6. Table 6 shows that by applying existing approaches on Example 1 we have $\tau_1(\check{F},\check{G}) = 0$, $\tau_2(\check{F},\check{G}) = 0$, $\tau_3(\check{F},\check{G}) = 0$, $\tau_4(\check{F},\check{G}) = 0$, $\tau_5(\check{F},\check{G}) = 0$ and $\tau_6(\check{F},\check{G}) = 1$. It is clear by observation that there is a mere correlation between F and G, but all the existing approaches give misinformation that there is no correlation. So, other approaches are not suitable in this context. 6th approach τ_6 gives the result that there is perfect correlation by giving the resulting value equal to 1, which is again a wrong result as $\check{F}$ ‘ $\check{G}$. In Example 2, it is visible that τ_1, τ_3, τ_4 are unable to calculate the result as they are giving values > 1. So, they do not satisfy the basic definition of CC that its value is always f 1. Example 3 shows that τ_4 gives output 0.67 ‘ 1 for two equal sets, τ_6 and τ_7 give $\frac{0}{0}$ forms. Therefore, τ_4 does not follow the basic definition of correlation coefficient that there is always a perfect correlation between two equal sets. τ_6 and τ_7 are unable to give the result when sets are equal. Again, in Example 4, after the application of the new approach we get $\tau_J(\check{F},\check{G}) = 0.376$, while other approaches give results $\tau_6(\check{F},\check{G}) = 1$ and $\tau_7(\check{F},\check{G}) = 1$. Here $\check{F}$ ‘ $\check{G}$, but these two approaches give misinformation that these sets have a perfect correlation between them. Here, the new approach gives the correct result. Of course, there is a correlation between these sets, but not that strong. So, in this case, other existing approaches do not suit well.

Finally, when the new approach is applied to Example 1, we get $\tau_J(\check{F},\check{G}) = 0.483$, which shows there is a weak correlation between the sets, rather than there is no correlation. In Example 2, we have $\tau_J(\check{F},\check{G}) = 0.765$, while other approaches are giving values > 1 and Example 3 give $\tau_J(F,G) = 1$, which shows that the equal sets always have perfect correlation, on the other hand existing approaches do not show a perfect correlation in this case. In Example 5, we get the output $\tau_J(\check{F},\check{G}) = 0.748$. By applying one of the existing approaches, we get $\tau_8(\check{F},\check{G}) = 0.633$. So, it is clear that the new approach of finding CC between $IFSs$ is the most reliable among all the existing approaches, as it satisfies the basic definition of CC. Its rate of performance is very good. It considers the hesitation index also, so its advantages are visible through examples, also its performance rate is the highest among all.

Table 5
Correlation Coefficient by new approach for different values of α.

α	ϕ_1	ϕ_2	ϕ_3	φ_1	φ_2	φ_3	ψ_1	ψ_2	ψ_3	$\tau_J(\check{F},\check{G})$
3	0.909	0.967	1.000	0.950	0.929	1.000	0.977	0.961	1.000	0.826
4	0.935	0.976	1.000	0.963	0.947	1.000	0.983	0.971	1.000	0.833
5	0.949	0.981	1.000	0.971	0.958	1.000	0.986	0.977	1.000	0.837
6	0.959	0.985	1.000	0.976	0.965	1.000	0.989	0.981	1.000	0.840
7	0.965	0.987	1.000	0.979	0.971	1.000	0.990	0.984	1.000	0.842
8	0.969	0.989	1.000	0.982	0.974	1.000	0.992	0.986	1.000	0.843
9	0.973	0.99	1.000	0.984	0.977	1.000	0.993	0.987	1.000	0.844
10	0.976	0.991	1.000	0.986	0.980	1.000	0.993	0.989	1.000	0.845

Table 6
Comparison between existing and proposed approach.

Approach	*Example* 1	*Example* 2	*Example* 3	*Example* 4	*Example* 5
τ_1 [20]	**0**	NA	1.000	0.699	0.999
τ_2 [21]	**0**	**0**	1.000	0.921	0.704
τ_3 [25]	**0**	NA	**0.662**	0.615	0.644
τ_4 [25]	**0**	NA	**0.67**	0.551	0.0,471
τ_5 [25]	**0**	**0**	1.000	0.921	0.704
τ_6 [26]	**1.000**	0.774	$\frac{0}{0}$ **form**	**1.000**	0.900
τ_7 [27]	0.666	0.77	$\frac{0}{0}$ **form**	**1.000**	0.780
τ_8 [28]	0.25	0.554	1.000	**1.000**	0.633
Proposed	0.483	0.765	1.000	0.376	0.748

4.1.1. Some properties of the new approach for correlation coefficient of IFSs

Here, the validity of the new approach to a CC is shown with the help of some properties:

Theorem 1. *Let $\check{F}$ and $\check{G}$ be two $IFSs$, then the new approach of CC satisfies the following properties:*

1. $\tau_J(\check{F},\check{G}) = \tau_J(\check{G},\check{F})$
2. $0 \le \tau_J(\check{F},\check{G}) \le 1$
3. $\tau_J(\check{F},\check{G}) = 1$ *iff* $\check{F} = \check{G}$

Proof.

1. We know that

$$\tau_J(\check{F},\check{G}) = \frac{1}{3N}\sum_{i=1}^{N}\left(\phi_i(1-\Delta m_i)+\varphi_i(1-\Delta n_i)+\psi_i(1-\Delta\rho_i)\right),$$

Then

$$\begin{aligned}\tau_J(\check{F},\check{G}) = \frac{1}{3N}\sum_{i=1}^{N}\Bigg(&\phi_i\left(1-\frac{|m_{\check{F}}(u_i)-m_{\check{G}}(u_i)|+|m^2_{\check{F}}(u_i)-m^2_{\check{G}}(u_i)|}{2}\right)\\&+\varphi_i\left(1-\frac{|n_{\check{F}}(u_i)-n_{\check{G}}(u_i)|+|n^2_{\check{F}}(u_i)-n^2_{\check{G}}(u_i)|}{2}\right)\\&+\psi_i\left(1-\frac{|\rho_{\check{F}}(u_i)-\rho_{\check{G}}(u_i)|+|\rho^2_{\check{F}}(u_i)-\rho^2_{\check{G}}(u_i)|}{2}\right)\Bigg)\\ = \frac{1}{3N}\sum_{i=1}^{N}\Bigg(&\phi_i\left(1-\frac{|m_{\check{G}}(u_i)-m_{\check{F}}(u_i)|+|m^2_{\check{G}}(u_i)-m^2_{\check{F}}(u_i)|}{2}\right)\\&+\varphi_i\left(1-\frac{|n_{\check{G}}(u_i)-n_{\check{F}}(u_i)|+|n^2_{\check{G}}(u_i)-n^2_{\check{F}}(u_i)|}{2}\right)\\&+\psi_i\left(1-\frac{|\rho_{\check{G}}(u_i)-\rho_{\check{F}}(u_i)|+|\rho^2_{\check{G}}(u_i)-\rho^2_{\check{F}}(u_i)|}{2}\right)\Bigg)\end{aligned}$$

$= \tau_J(\check{F},\check{G})$

2. It is clear from the formation of $\tau_J(\check{F},\check{G})$ that $\tau_J(\check{F},\check{G}) \ge 0$. Now, we just have to show that $\tau_J(\check{F},\check{G}) \le 1$. Let us assume that

$$\sum_{i=1}^{N}\phi_i(1-\Delta m_i) = \zeta_1,\quad \sum_{i=1}^{N}\varphi_i(1-\Delta n_i) = \zeta_2,\quad \sum_{i=1}^{N}\psi_i(1-\Delta\rho_i) = \zeta_3,$$

By using Cauchy–Schwarz Inequality, we get

$$\tau_J(\check{F},\check{G}) = \frac{1}{3N}\sum_{i=1}^{N}\phi_i(1-\Delta m_i)+\varphi_i(1-\Delta n_i)+\psi_i(1-\Delta\rho_i), \text{ where}$$

$$\le \frac{\sum_{i=1}^{N}\phi_i(1-\Delta m_i) + \sum_{i=1}^{N}\varphi_i(1-\Delta n_i) + \sum_{i=1}^{N}\psi_i(1-\Delta\rho_i)}{3N}$$

$$= \frac{\zeta_1+\zeta_2+\zeta_3}{3N}$$

Hence,

$$\begin{aligned}\tau_J(\check{F},\check{G}) - 1 &= \frac{\zeta_1+\zeta_2+\zeta_3}{3N} - 1 = \frac{\zeta_1+\zeta_2+\zeta_3-3N}{3N}\\&= -\frac{(3N-\zeta_1-\zeta_2-\zeta_3)}{3N} \le 0,\end{aligned}$$

Therefore, we have $\tau_J(\check{F},\check{G}) \le 1$. Hence, $\tau_J(\check{F},\check{G}) \in [0,1]$

3. Suppose $\check{F} = \check{G}$. Then, we have

$$|n_{\check{F}}(u_i) - n_{\check{G}}(u_i)| = |m_{\check{F}}(u_i) - m_{\check{G}}(u_i)| = |\rho_{\check{F}}(u_i) - \rho_{\check{G}}(u_i)| = 0\ \forall i.$$

So, we get

$$\Delta m_i = \Delta n_i = \Delta\rho_i = 0;\ \Delta m_{min} = \Delta n_{min} = \Delta\rho_{min} = \Delta m_{max} = \Delta n_{max} = \Delta\rho_{max} = 0;$$

Also $\phi_i = \varphi_i = \psi_i = 1\ \forall i$.

Hence $\tau_J(\check{F},\check{G}) = \frac{1}{3N}\sum_{i=1}^{N} 3N = 1$.

Conversely, if $\tau_J(\check{F},\check{G}) = 1$, then $\check{F} = \check{G}$.

5. Applications of the new approach of correlation coefficient

Here, we explain the relevance of the new CC approach in different fields. It can be used to detect different kinds of patterns. Pattern detection is the art of categorizing patterns based on a machine learning algorithm. Pattern detection has to do with the grouping of data based on already gained knowledge to aid inference. In most cases, the art of pattern detection is enmeshed with uncertainties, which justifies the use of the soft computing approach of the Intuitionistic Fuzzy CC technique. Also, this new approach can be applied in the field of medical science to diagnose the diseases of patients. The highest correlation of the symptoms of patients with the diseases gives the correct results about the patient's illness. This approach is also useful in the context of decision-making issues. It works as a helpful aid in finding the best alternative among so many alternatives. It is reliable in handling $MADM$ and $MCDM$ issues.

To show the applications of the new approach of CC for $IFSs$ in detail, we have done some experiments which are explained in the following subsections.

5.1. Pattern detection

let us suppose that there are known patterns within a sample space and an unknown pattern within the same space that needs to be grouped into any of the similar known patterns using the new approach. The highest CC determines the classification of that unknown sample.

Table 7
Classification of medicines in the form of Intuitionistic fuzzy values.

Patterns	l_1	l_2	l_3	l_4	l_5
$m_{\check{S}_1}$	0.578	0.055	0.202	0.414	0.025
$n_{\check{S}_1}$	0.322	0.888	0.598	0.355	0.350
ρ_{S_1}	0.100	0.057	0.200	0.231	0.625
$m_{\check{S}_2}$	0.135	0.035	0.050	0.200	0.115
$n_{\check{S}_2}$	0.635	0.820	0.700	0.575	0.815
ρ_{S_2}	0.230	0.145	0.250	0.225	0.070
$m_{\check{S}_3}$	0.445	0.662	1.000	0.325	0.485
$n_{\check{S}_3}$	0.365	0.298	0.000	0.535	0.305
ρ_{S_3}	0.190	0.040	0.000	0.040	0.210
$m_{\check{S}_4}$	0.000	0.626	0.546	0.479	1.000
$n_{\check{S}_4}$	0.000	0.314	0.424	0.321	0.000
ρ_{S_4}	0.000	0.060	0.030	0.200	0.000
$m_{\check{J}}$	0.529	0.526	0.210	0.035	0.538
$n_{\check{J}}$	0.203	0.334	0.675	0.555	0.432
$\rho_{\check{J}}$	0.262	0.140	0.115	0.410	0.030

Table 8
Results obtained by New approach and existing approaches.

Approaches	$(\check{S}_1, \check{J})$	$(\check{S}_2, \check{J})$	$(\check{S}_3, \check{J})$	$(\check{S}_4, \check{J})$	*Identified category*
$New\ Approach = \tau_J$	0.758	0.765	0.788	0.702	$\check{S}_3$
τ_8 [28]	0.614	0.654	0.742	0.575	$\check{S}_3$
τ_7 [27]	0.782	0.772	0.826	0.720	$\check{S}_3$
τ_6 [26]	0.737	0.775	0.814	0.738	$\check{S}_3$
τ_5 [25]	0.704	0.758	0.711	0.677	$\check{S}_2$
τ_4 [25]	NA	NA	NA	NA	NA
τ_3 [25]	NA	NA	NA	NA	NA
τ_1 [20]	NA	NA	NA	NA	NA

Example 7. Suppose that we have a case of pattern detection of some medicines. Four categories of medicine are given, which contain five types of chemicals. There is a typical category of hybrid medicine also. Here, four categories of medicine are represented by $IFSs\ \check{S}_1 = Antiseptics, \check{S}_2 = Antipyretics, \check{S}_3 = Antibiotics, \check{S}_4 = Anticonvulsant$ in the feature space $L = \{l_1 = Penicillin, l_2 = Gentamycin, l_3 = Furacin, l_4 = Terpineol, l_5 = Chloroxylenol\}$ containing different chemicals. Assume that there is an unclassified category of hybrid medicine $\check{J}$, then we find out to which category this unclassified medicine can belong. Intuitionistic fuzzy values of given medicines and unclassified medicine are provided in Table 7:

After applying the new approach and existing approaches, we get the output provided in the form of Table 8.

After observing the data in Table 8, it is clear that the new approach for finding the CC is capable of detecting the pattern. Also, some approaches $\tau_1, \tau_3\ and\ \tau_4$ fail to detect the pattern. The result obtained by new as well as existing approaches is the same. The unknown hybrid medicine $\check{J}$ belongs to $\check{S}_3 = Antibiotics$ category of medicine.

5.2. Medical diagnosis

In this section, we use our new approach of correlation in the field of medical science for medical diagnosis purposes as follows:

Example 8. Let us consider a data set containing four patients $\tilde{Q} = \{Elena, Marco, Ethan, Devin\}$, a data set containing symptoms of these patients $\tilde{X} = \{x_1 = Dizziness, x_2 = Sweating, x_3 = Fainting, x_4 = Red\ Skin, x_5 = Pulse\ rate\}$ and a data set containing five illnesses $\tilde{Y} = \{y_1 = Heat\ Stroke, y_2 = Heat\ Exhaustion, y_3 = Heat\ Cramp, y_4 = Sunburn, y_5 = Heat\ Rash\}$. All these sets are $IFSs$. The relationship between patients and symptoms is represented by Table 9 and the relationship between symptoms and illnesses is represented by Table 10. In this example, our target is to diagnose the illness of the patients. Here, we apply a new approach of CC as well as some of the existing approaches and finally compare the results. The result calculated by the new approach is represented in Table 11.

Table 9
Data of Patients having symptoms in the form of $IFSs$.

Patient	x_1	x_2	x_3	x_4	x_5
Elena	(0.8,0.1,0.1)	(0.2,0.6,0.2)	(0.3,0.5,0.2)	(0.6,0.1,0.3)	(0.8,0.2,0.0)
Marco	(0.6,0.3,0.1)	(0.4,0.3,0.3)	(0.4,0.5,0.1)	(0.5,0.3,0.2)	(0.3,0.6,0.1)
Ethan	(0.5,0.4,0.1)	(0.6,0.2,0.2)	(0.7,0.1,0.2)	(0.5,0.2,0.3)	(0.4,0.4,0.2)
Devin	(0.3,0.4,0.3)	(0.5,0.2,0.3)	(0.7,0.2,0.1)	(0.3,0.3,0.4)	(0.6,0.2,0.2)

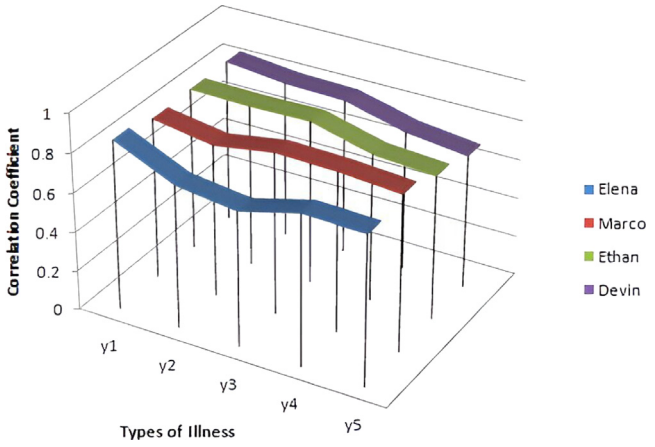

Fig. 1. Results obtained by new approach for medical diagnosis.

Also, the results calculated by the existing approaches namely [20, 21,25,28] are represented in the form of Table 12, Table 13, Table 14, and Table 15 respectively.

Two approaches of [25] namely τ_3, τ_4 are unable to identify the illness. Therefore, the consolidated result is represented in the form of Table 16. It clearly depicts that *Elena*, *Marco*, *Ethan* and *Devin* are suffering from *Heat Stroke*, *Heat Cramp*, *Heat Cramp* and *Heat Stroke* respectively (see Fig. 1).

5.3. Clustering in $IFSs$

In this section, we use our new approach of CC in clustering $IFSs$. The algorithm to solve such problems of clustering is as follows:

Requirement: $\tilde{Q} = \{Q_1, Q_2, \ldots, Q_r\}$, the set of $IFSs$ on $\ddot{U} = \{u_1, u_2, \ldots, u_n\}$.

STEP-I Calculate $\tau_J(Q_i, Q_j)$ using Eqs. (16), (17), (18), and (19). Use them and Definition 5 to create an association matrix $\tilde{Y} = [y_{ij}]_{r\times r}$, where i,j=1,2,...,r.

STEP-II Now, construct composition matrix $\tilde{Y}^{2^{k+1}} = \tilde{Y}^{2^k} \acute{y} \tilde{Y}^{2^k}, k = 1, 2, \ldots$ using Definition 6. Stop until $\tilde{Y}^{2^{k+1}} = \tilde{Y}^{2^k}$.

STEP-III Now, by using Definition 7 construct a λ- cutting matrix $\tilde{Y}_\lambda$ of matrix $\tilde{Y}^{2^k}$ for a confidence level $\lambda \in [0, 1]$.

STEP-IV Now, put all those $IFSs\ Q_i, Q_j$ in the same class whose ith and jth column are the same in λ- cutting matrix $\tilde{Y}_\lambda$. The whole process is explained with the help of the following example:

Example 9 ([28]). Let $\tilde{Q} = \{Q_1, Q_2, Q_3, Q_4, Q_5\}$ be the set of $IFSs$ on $\ddot{U} = \{u_1, u_2, u_3, u_4, u_5, u_6\}$ as follows:

Q_1 = {(0.2, 0.4, 0.4), (0.4, 0.1, 0.5), (0.9, 0.0, 0.1), (0.8, 0.1, 0.1), (0.2, 0.5, 0.3), (0.7, 0.1, 0.2)},

Q_2 = {(0.6, 0.3, 0.1), (0.5, 0.2, 0.3), (0.6, 0.1, 0.3), (0.7, 0.1, 0.2), (0.3, 0.6, 0.1), (0.4, 0.3, 0.3)},

Q_3 = {(0.5, 0.2, 0.3), (0.3, 0.6, 0.1), (0.6, 0.3, 0.1), (0.7, 0.1, 0.2), (0.6, 0.2, 0.2), (0.5, 0.3, 0.2)},

Q_4 = {(0.3, 0.5, 0.2), (0.6, 0.1, 0.3), (0.4, 0.3, 0.3), (0.8, 0.1, 0.1), (0.1, 0.6, 0.3), (0.5, 0.4, 0.1)},

Q_5 = {(0.4, 0.4, 0.2), (0.8, 0.1, .1), (0.5, 0.1, 0.4), (0.6, 0.2, 0.2), (0.4, 0.5, 0.1), (0.3, 0.2, 0.5)}.

Table 10
Data of Symptoms to diagnose the Illness in the form of $IFSs$.

Illness	x_1	x_2	x_3	x_4	x_5
Heat Stroke	(0.7,0.2,0.1)	(0.3,0.6,0.1)	(0.6,0.3,0.1)	(0.5,0.2,0.1)	(0.7,0.1,0.2)
Heat Exhaustion	(0.8,0.2,0.0)	(0.8,0.1,0.1)	(0.7,0.2,0.1)	(0.2,0.7,0.1)	(0.5,0.3,0.2)
Heat Cramp	(0.2,0.7,0.1)	(0.6,0.2,0.2)	(0.3,0.5,0.2)	(0.3,0.6,0.1)	(0.4,0.4,0.2)
Sunburn	(0.1,0.8,0.1)	(0.3,0.5,0.2)	(0.2,0.6,0.2)	(0.8,0.1,0.1)	(0.3,0.6,0.1)
Heat Rash	(0.2,0.8,0.0)	(0.2,0.7,0.1)	(0.2,0.5,0.3)	(0.7,0.2,0.1)	(0.2,0.7,0.1)

Table 11
Results calculated by new correlation coefficient approach τ_J.

Patient	y_1	y_2	y_3	y_4	y_5
Elena	**0.866**	0.728	0.689	0.771	0.761
Marco	0.825	0.776	**0.826**	0.818	0.796
Ethan	0.833	0.827	**0.834**	0.753	0.731
Devin	**0.838**	0.805	0.797	0.718	0.681

Table 12
Results calculated by existing correlation coefficient approach τ_1.

Patient	y_1	y_2	y_3	y_4	y_5
Elena	**0.958**	0.713	0.650	0.702	0.700
Marco	0.852	0.839	**0.859**	0.842	0.848
Ethan	0.867	**0.895**	0.824	0.773	0.730
Devin	0.883	**0.895**	0.861	0.705	0.676

Table 13
Results calculated by existing correlation coefficient approach τ_2.

Patient	y_1	y_2	y_3	y_4	y_5
Elena	**0.948**	0.705	0.657	0.703	0.702
Marco	0.849	0.829	**0.858**	0.839	0.834
Ethan	0.866	**0.874**	0.821	0.763	0.725
Devin	**0.871**	0.845	0.839	0.688	0.645

Table 14
Results calculated by existing correlation coefficient approach τ_5.

Patient	y_1	y_2	y_3	y_4	y_5
Elena	**0.949**	0.705	0.657	0.703	0.702
Marco	0.809	0.829	**0.859**	0.840	0.834
Ethan	0.865	**0.875**	0.821	0.763	0.724
Devin	**0.872**	0.846	0.840	0.689	0.646

Table 15
Results calculated by existing correlation coefficient approach τ_8.

Patient	y_1	y_2	y_3	y_4	y_5
Elena	**0.852**	0.608	0.586	0.507	0.665
Marco	0.624	0.698	**0.760**	0.702	0.628
Ethan	0.737	0.746	**0.750**	0.638	0.628
Devin	**0.786**	0.772	0.758	0.626	0.591

Table 16
Consolidated result of medical diagnosis.

Patient	τ_J	τ_1 [20]	τ_2 [21]	τ_3 [25]	τ_4 [25]	τ_5 [25]	τ_8 [28]
Elena	y_1	y_1	y_1	NA	NA	y_1	y_1
Marco	y_3	y_3	y_3	NA	NA	y_3	y_3
Ethan	y_3	y_2	y_2	NA	NA	y_2	y_3
Devin	y_1	y_2	y_1	NA	NA	y_1	y_1

STEP-I By using Eqs. (16), (17), and (18), we get association matrix as follows:

$$\tilde{Y} = \begin{bmatrix} 1.000 & 0.803 & 0.796 & 0.837 & 0.748 \\ 0.803 & 1.000 & 0.846 & 0.879 & 0.878 \\ 0.796 & 0.846 & 1.000 & 0.801 & 0.796 \\ 0.837 & 0.879 & 0.801 & 1.000 & 0.838 \\ 0.748 & 0.878 & 0.796 & 0.838 & 1.000 \end{bmatrix};$$

STEP-II Now construct the composition matrix:

$$\tilde{Y}^2 = \tilde{Y} \acute{y} \tilde{Y} = \begin{bmatrix} 1.000 & 0.837 & 0.803 & 0.837 & 0.837 \\ 0.837 & 1.000 & 0.846 & 0.879 & 0.878 \\ 0.803 & 0.846 & 1.000 & 0.846 & 0.846 \\ 0.837 & 0.879 & 0.846 & 1.000 & 0.878 \\ 0.837 & 0.878 & 0.846 & 0.878 & 1.000 \end{bmatrix};$$

$$\tilde{Y}^4 = \tilde{Y}^2 \acute{y} \tilde{Y}^2 = \begin{bmatrix} 1.000 & 0.837 & 0.837 & 0.837 & 0.837 \\ 0.837 & 1.000 & 0.846 & 0.879 & 0.878 \\ 0.837 & 0.846 & 1.000 & 0.846 & 0.846 \\ 0.837 & 0.879 & 0.846 & 1.000 & 0.878 \\ 0.837 & 0.878 & 0.846 & 0.878 & 1.000 \end{bmatrix};$$

and

$$\tilde{Y}^8 = \tilde{Y}^4 \acute{y} \tilde{Y}^4 = \begin{bmatrix} 1.000 & 0.837 & 0.837 & 0.837 & 0.837 \\ 0.837 & 1.000 & 0.846 & 0.879 & 0.878 \\ 0.837 & 0.846 & 1.000 & 0.846 & 0.846 \\ 0.837 & 0.879 & 0.846 & 1.000 & 0.878 \\ 0.837 & 0.878 & 0.846 & 0.878 & 1.000 \end{bmatrix};$$

It is clear that $\tilde{Y}^8 = \tilde{Y}^4$. Therefore, $\tilde{Y}^4$ is the equivalence matrix.

Step-III Now, we construct λ- cutting matrix $\tilde{Y}_\lambda$ for different possible values of λ. The possible λ- cutting matrices $\tilde{Y}_\lambda$ are as follows:

1. If $0 \leq \lambda \leq 0.837$, then

$$\tilde{Y}_\lambda = \begin{bmatrix} 1.000 & 1.000 & 1.000 & 1.000 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 & 1.000 \\ 1.000 & 1.000 & 1.000 & 1.000 & 1.000 \end{bmatrix};$$

2. If $0.837 < \lambda \leq 0.846$, then

$$\tilde{Y}_\lambda = \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 1.000 & 1.000 & 1.000 \\ 0.000 & 1.000 & 1.000 & 1.000 & 1.000 \\ 0.000 & 1.000 & 1.000 & 1.000 & 1.000 \\ 0.000 & 1.000 & 1.000 & 1.000 & 1.000 \end{bmatrix};$$

3. If $0.846 < \lambda \leq 0.878$, then

$$\tilde{Y}_\lambda = \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 & 1.000 & 1.000 \\ 0.000 & 0.000 & 1.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 & 1.000 & 1.000 \\ 0.000 & 1.000 & 0.000 & 1.000 & 1.000 \end{bmatrix};$$

4. If $0.878 < \lambda \leq 0.879$, then

$$\tilde{Y}_\lambda = \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 & 1.000 & 0.000 \\ 0.000 & 0.000 & 1.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 & 1.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 1.000 \end{bmatrix};$$

5. If $0.879 < \lambda \leq 1.000$, then

$$\tilde{Y}_\lambda = \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 1.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 1.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 1.000 \end{bmatrix};$$

STEP-IV Now, we get all the possible classifications of $\{Q_1, Q_2, Q_3, Q_4, Q_5\}$ according to possible λ- cutting matrices $\tilde{Y}_\lambda$ mentioned in STEP-III.

1. If $0 \leq \lambda \leq 0.837$, then we have one cluster: $\{Q_1, Q_2, Q_3, Q_4, Q_5\}$.
2. If $0.837 < \lambda \leq 0.846$, then we have two clusters: $\{Q_1\}$ and $\{Q_2, Q_3, Q_4, Q_5\}$.
3. If $0.846 < \lambda \leq 0.878$, then we have three clusters: $\{Q_1\}$, $\{Q_2, Q_4, Q_5\}$ and $\{Q_3\}$.
4. If $0.878 < \lambda \leq 0.879$, then we have four clusters: $\{Q_1\}$, $\{Q_2, Q_4\}$, $\{Q_3\}$ and $\{Q_5\}$.
5. If $0.879 < \lambda \leq 1.000$, then we have five clusters: $\{Q_1\}$, $\{Q_2\}$, $\{Q_3\}$, $\{Q_4\}$ and $\{Q_5\}$.

After analyzing the above example [28], it is clear that the new approach of the CC is effectively applicable in clustering issues of $IFSs$. By using the new approach, we get five possibilities, on the other hand in [21] only three possibilities were found. So, it can be observed that the new approach is an approach with high accuracy in clustering.

5.4. Proposed VIKOR method based on correlation coefficient of IFS

In the VIKOR method, only group utility values and individual regret that are near to the positive ideal are considered and then, we select only those values for alternatives that are near to the positive ideal. In the TOPSIS method, those values are selected which are near to the positive ideal and distant from the negative ideal. Here, we propose an approach based on CC, in which values that are nearest to the positive ideal and most distant from the negative ideal are selected, and the relative closeness concept of TOPSIS is used. Steps to solve $MCDM$ issues by using the TOPSIS-VIKOR method:

Basic Requirements:

(i) Collection of alternatives.
(ii) Collection of Criteria.
(iii) Collection of Resource persons.

First of all, suppose that r alternatives are evaluated by t resource persons, which are based on s criteria; where alternatives belong to the set $J = \{J_1, J_2, \ldots, J_r\}$ and criteria belong to the set $K = \{K_1, K_2, \ldots, K_s\}$. Let $P = (p_{ij})_{r\times s} (1 \leq i \leq r, 1 \leq j \leq s)$ be the IF decision matrix, determined by using the results obtained by the expert evaluation as follows:

$$P = (p_{ij})_{r\times s} = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1s} \\ p_{21} & p_{22} & \cdots & p_{2s} \\ \vdots & \vdots & \ddots & \vdots \\ p_{r1} & p_{r2} & \cdots & p_{rs} \end{bmatrix};$$

where $p_{ij} = (m_{ij}, n_{ij}, \rho_{ij})$ is obtained by IF values. m_{ij}, n_{ij} *and* ρ_{ij} represent DoM, $DoNM$ and the hesitancy degree of the alternative J_i satisfies criterion K_j. Alternatives and criteria are represented by rows and columns of matrix P respectively.

STEP-I

Construct the normalized IF decision matrix, based on the responses provided by the resource persons regarding criteria and alternatives. Decision matrix P contains benefit and cost criteria. We convert the cost criteria into the benefit criteria. Now the normalized IF decision matrix is

$$B = (b_{ij})_{r\times s} = \begin{cases} b_{ij} = (m_{ij}, n_{ij}, \rho_{ij}), K_j \text{ is a benefit criterion}; \\ b_{ij}^C = (n_{ij}, m_{ij}, \rho_{ij}), K_j \text{ is a cost criterion}. \end{cases}$$

STEP-II

CASE-I (For unknown criteria weights)

Now determine the weights of the criteria. They are calculated by the following formula:

$$t_j = \frac{1 - I_j}{s - \sum_{j=1}^{s} I_j}, \text{where } t_j \in [0,1], \text{and } \sum_{j=1}^{s} t_j = 1. \tag{19}$$

I_j [59] is IF entropy of criterion K_j :

$$I_j = 1 - \frac{1}{2r} \sum_{i=1}^{r} (|m_{ij} - n_{ij}| + |m_{ij} - \rho_{ij}| + |n_{ij} - \rho_{ij}|). \tag{20}$$

Entropy given by (20) satisfies the properties of Definition 8. It is used to represent fuzziness of sets in the form of mathematical values. It gives information about the fuzzy sets. It is also used to calculate the criterion weight. Now, the process of calculating the criterion weight is illustrated with the help of an example. Let us suppose that we have 3 criteria and 5 alternatives. The decision matrix is as follows:

$$O = \begin{bmatrix} (0.3,0.4,0.3) & (0.6,0.3,0.1) & (0.4,0.4,0.2) \\ (0.4,0.3,0.3) & (0.5,0.4,0.1) & (0.4,0.5,0.1) \\ (0.7,0.1,0.2) & (0.3,0.5,0.2) & (0.6,0.1,0.3) \\ (0.1,0.3,0.6) & (0.6,0.2,0.2) & (0.9,0.1,0.0) \\ (0.7,0.2,0.1) & (0.5,0.1,0.4) & (0.3,0.5,0.2) \end{bmatrix};$$

Here by using (20) we get information about 3 criteria 0.62, 0.60, and 0.54 respectively. By using (19) we get weights of 3 criteria 0.306, 0.322, and 0.370 respectively. This shows that the weight of a criterion will decrease as we gain more information about it.

CASE-II (For partially known criteria weights)

Due to some kind of circumstances, sometimes resource persons are not able to assign numeric values to each criterion. In that case, they give their opinion in the form of intervals. Then, we compile the provided information in set $\bar{I}$. Then, criteria weights are calculated by the following formula:

$$P_{max} = \sum_{j=1}^{s} \tilde{t}_j \left(1 - \frac{1}{2r} \sum_{i=1}^{r} (|m_{ij} - n_{ij}| + |m_{ij} - \rho_{ij}| + |n_{ij} - \rho_{ij}|)\right)$$

where $\tilde{t}_j \in \bar{I}$ and $\sum_{j=1}^{s} \tilde{t}_j = 1$.

STEP-III

Here, we will find the positive ideal as well as negative ideal vectors corresponding to each criterion. U^+ *and* U^- are positive ideal value IF vectors and negative ideal value IF vectors respectively.

$$U^+ = [u_1^+ \quad u_2^+ \quad \cdots \quad u_s^+],$$

$$U^- = [u_1^- \quad u_2^- \quad \cdots \quad u_s^-],$$

where $u_j^+ = \max_i m_{ij}, \min_i n_{ij}, \min_i \rho_{ij}$ and

$$u_j^- = \min_i m_{ij}, \max_i n_{ij}, \min_i \rho_{ij}. \tag{21}$$

STEP-IV

Now, determine the positive ideal correlation coefficient matrix V^+ and the negative ideal correlation coefficient matrix V^-. Then, calculate the CC of U^+ and B under each criterion, as well as the CC of U^- and B under each criterion.

$$V^+ = (v_{ij}^+)_{r\times s}, V^- = (v_{ij}^-)_{r\times s},$$

where $v_{ij}^+ = \tau_J(b_{ij}, u_j^+), v_{ij}^- = \tau_J(b_{ij}, u_j^-)$. (22)

STEP-V

Obtain the vectors M^+ *and* M^-, nearest to and most distant from the positive ideal for all criteria. Obtain the vectors N^+ *and* N^-, nearest to and most distant from the negative ideal for all criteria.

$$M^+ = [m_1^+ \quad m_2^+ \quad \cdots \quad m_s^+], M^- = [m_1^- \quad m_2^- \quad \cdots \quad m_s^-];$$

$$N^+ = [n_1^+ \quad n_2^+ \quad \cdots \quad n_s^+], N^- = [n_1^- \quad n_2^- \quad \cdots \quad n_s^-];$$

Fig. 2. Framework of suggested $MCDM$ issue.

where $m_j^+ = \max_i \ v_{ij}^+, \ \ m_j^- = \min_i \ v_{ij}^+ \ ; \ \ n_j^+ = \max_i \ v_{ij}^-, \ \ n_j^- = \min_i \ v_{ij}^- \ .$

(23)

STEP-VI

Determine the normalized nearest positive ideal group utility value MS_i and normalized nearest positive ideal individual regret value MR_i of alternative J_i.

$$MS_i = \sum_{j=1} t_j \frac{m_j^+ - v_{ij}^+}{m_j^+ - m_j^-} \ , \quad MR_i = \max_j \ t_j \frac{p_j^+ - h_{ij}^+}{m_j^+ - m_j^-} \ . \tag{24}$$

Also, determine the normalized nearest negative ideal group utility value NS_i and the normalized nearest negative ideal individual regret value NR_i of alternative J_i.

$$NS_i = \sum_{j=1} t_j \frac{n_j^+ - v_{ij}^-}{n_j^+ - n_j^-} \ , \quad NR_i = \max_j \ t_j \frac{n_j^+ - v_{ij}^-}{n_j^+ - n_j^-} \ . \tag{25}$$

STEP-VII

Calculate the nearest positive ideal VIKOR index Z_i^+ and the nearest negative ideal VIKOR index Z_i^- of alternative J_i.

$$Z_i^+ = \delta \frac{MS_i - MS^*}{MS^- - MS^*} + (1-\delta)\frac{MR_i - MR^*}{MR^- - MR^*}, \tag{26}$$

$$Z_i^- = \delta \frac{NS_i - NS^*}{NS^- - NS^*} + (1-\delta)\frac{NR_i - NR^*}{NR^- - NR^*}, \tag{27}$$

where $MS^- = \max_i \ MS_i, MS^* = \min_i \ MS_i,$

$$MR^- = \max_i \ MR_i, MR^* = \min_i \ MR_i; \tag{28}$$

$$NS^- = \max_i \ NS_i, NS^* = \min_i \ NS_i, NR^- = \max_i \ NR_i, NR^* = \min_i \ NR_i. \tag{29}$$

Here, δ and $1-\delta$ represent the weight of the majority of criteria and the individual regret respectively. Generally, a value at $\delta = 0.5$ is considered.

STEP-VIII

Now evaluate the relative closeness coefficient BZ_i of alternative J_i,

$$BZ_i = \frac{Z_i^+}{Z_i^+ + Z_i^-}. \tag{30}$$

When the degree of relative closeness BZ_i decreases, the preference order of alternative J_i, 1 f i f r improves.

Experimental results

Now, we use numerical examples to see the effective use of the TOPSIS-VIKOR method in $MCDM$ problems. Nowadays, inorganic fertilizers are in demand in the field of agriculture and farming, since they act fast, and they are economical, they provide the exact amount of required nutrients to the soil for the growth of plants as all farms have different requirements based on their soil. To help in these types of decision-making issues, we have taken a case study of finding the best inorganic fertilizers among the shortlisted five fertilizers by using the VIKOR-TOPSIS method. The Framework of the numerical example is shown with the help of Fig. 2.

Example 10. Suppose, we want to choose the best inorganic fertilizer among five alternatives $J_1 = Ammonium\ Nitrate$, $J_2 = Ammonium\ Sulphate$, $J_3 = Aqua\ Ammonia$, $J_4 = Potassium\ Nitrate$, $J_5 = Sodium\ Nitrate$, represented by the set $J = \{J_1, J_2, J_3, J_4, J_5\}$, based on five criteria $K_1 = Solubility$, $K_2 = Particle\ size$, $K_3 = Salt\ Indices$, $K_4 = Economic$, $K_5 = Environment\ Friendly$, represented by the set $K = \{K_1, K_2, K_3, K_4, K_5\}$. Let the experts $\bar{R}_i$, (1 f i f 10) evaluate these alternatives based on the above-mentioned five criteria, their responses are shown in Table 17 and we obtain the IF decision matrix as follows:

$$P = \begin{array}{c} \\ J_1 \\ J_2 \\ J_3 \\ J_4 \\ J_5 \end{array} \begin{array}{c} \begin{array}{ccccc} K_1 & K_2 & K_3 & K_4 & K_5 \end{array} \\ \begin{bmatrix} (0.3,0.4,0.3) & (0.4,0.1,0.5) & (0.7,0.2,0.1) & (0.6,0.3,0.1) & (0.4,0.4,0.2) \\ (0.4,0.3,0.3) & (0.3,0.3,0.4) & (0.6,0.2,0.2) & (0.5,0.4,0.1) & (0.4,0.5,0.1) \\ (0.7,0.1,0.2) & (0.8,0.1,0.1) & (0.5,0.3,0.2) & (0.3,0.5,0.2) & (0.6,0.1,0.3) \\ (0.1,0.3,0.6) & (0.2,0.4,0.4) & (0.4,0.2,0.4) & (0.6,0.2,0.2) & (0.9,0.1,0.0) \\ (0.7,0.2,0.1) & (0.6,0.3,0.1) & (0.5,0.2,0.3) & (0.5,0.1,0.4) & (0.3,0.5,0.2) \end{bmatrix} \end{array};$$